

Penalty-Based No-Arbitrage Enforcement for Neural Volatility Surfaces Under Sparse Strike Data

Diego Urdaneta
M2 Research
diego@achong.com

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Abstract

Standard volatility surface calibration methods face a fundamental tradeoff: parametric models (SVI, SABR) guarantee no-arbitrage but fail under sparse or irregular strike grids; neural networks interpolate flexibly but routinely produce butterfly and calendar arbitrage. **We show this tradeoff can be made favorable, though not eliminated.**

We train a small MLP (3 layers, 64–32–16, Softplus output) with squared-hinge penalty losses on Breeden-Litzenberger density positivity, calendar monotonicity, and Roger-Lee wing bounds. This is textbook penalty-method constrained optimization applied to volatility surface fitting—the technique is not novel. The contribution is showing that this straightforward approach, combined with a warmup schedule and loss normalization, is numerically stable enough to work even when 50% of strikes are missing, which was not previously demonstrated.

Main result: On S&P 500 options with randomly dropped strikes (40–60% missing), our method achieves 22.7 bps RMSE at 40% dropout with exploitable arbitrage below 5×10^{-4} bps at evaluation grid points, while per-slice SVI calibration produces 1.4 bps of butterfly arbitrage profit and unconstrained neural networks produce 11.2 bps. Under full data, the accuracy gap versus unconstrained methods is only 4.3%. Calibration takes 89ms per surface on GPU (412ms end-to-end including data preprocessing).

What we do not claim: Global optimality, convexity of the admissible set, or theoretical characterization of required penalty weights. Our constraint enforcement is a penalty-based heuristic that works empirically, not a projection onto a convex set. Arbitrage-freeness is verified at discrete evaluation grid points; continuous inter-grid-point guarantees are not provided.

Keywords: Volatility Surfaces, No-Arbitrage Constraints, Sparse Data Interpolation, Constraint-Informed Learning

Acronyms: SVI = Stochastic Volatility Inspired; SABR = Stochastic Alpha Beta Rho; MLP = Multi-Layer Perceptron; CINN = Constraint-Informed Neural Network¹; EPP = Exploitable Profit Potential; RMSE = Root Mean Squared Error; bps = basis points (0.01%).

¹Not to be confused with Conditional Invertible Neural Networks, which share the same acronym in the generative modeling literature.

1 Introduction

1.1 The Problem: No-Arbitrage Under Sparsity

Implied volatility surfaces must satisfy stringent no-arbitrage conditions—positive density, calendar monotonicity, finite moment bounds—to be consistent with any risk-neutral measure. Violating these conditions is not merely aesthetically displeasing; it creates exploitable trading opportunities and corrupts downstream Greeks.

The standard approach uses parametric models: SVI [Gatheral, 2004], SSVI [Gatheral and Jacquier, 2014], or SABR [Hagan et al., 2002]. These guarantee arbitrage-freeness by construction but suffer a critical weakness:

Observation 1 (Per-Slice Parametric Brittleness). *For sparse strike grids (≤ 8 strikes per expiry, corresponding to 60% dropout from typical market data), standard per-slice SVI calibration—fitting 5 parameters independently per expiry using Levenberg-Marquardt with random initialization—produces butterfly arbitrage opportunities averaging 3.2 bps EPP due to underdetermined parameter identification. Specifically:*

- (a) *Wild extrapolations in the wings (insufficient curvature constraints)*
- (b) *Spurious local optima from underdetermined parameters*
- (c) *Sensitivity to noise that dwarfs the volatility signal*

Caveat: *Joint calibration methods mitigate but do not eliminate these pathologies. Our benchmark includes SSVI [Gatheral and Jacquier, 2014] with joint calibration across expiries; it still produces 2.14 bps EPP at 60% dropout (Table 7). Further improvements may be possible with quasi-explicit SVI [De Marco and Martini, 2009] or no-arbitrage parameter constraints [Martini and Mingone, 2022]. See Section 2 for a full discussion.*

The alternative—neural network interpolation—handles irregular grids naturally but ignores arbitrage constraints entirely. Standard MLPs achieve excellent in-sample fit but produce surfaces where 30–40% of grid points violate butterfly positivity.

The core question: Can we have both? Flexible interpolation under sparse data *and* guaranteed arbitrage-freeness?

1.2 Our Answer: Yes, With Caveats

We show that constraint-informed neural networks can achieve both, with caveats we state upfront:

1. We sacrifice 4–5% fitting accuracy relative to unconstrained methods
2. We require $\sim 1.7\times$ more calibration time than unconstrained MLPs (412ms vs 245ms)
3. We make no claims about optimality—our method finds *a* feasible solution, not *the* best one

In exchange, we get:

1. Near-zero exploitable arbitrage (< 0.05 bps) across all test conditions, including 50% strike dropout

2. Stable behavior under initialization variance (std of RMSE < 0.8 bps across 20 seeds)
3. No manual tuning per surface—a single penalty configuration works universally

1.3 Why This Matters

For practitioners: Sparse grids are the norm, not the exception. Single-stock options, emerging markets, and illiquid tenors routinely have 5–10 strikes per expiry. A method that degrades gracefully under sparsity is practically essential.

For researchers: The failure mode of soft-penalty methods is underappreciated. We show that “soft” arbitrage penalties ($\lambda \leq 0.1$) achieve 90%+ “arbitrage-free” points but leave 2–3 bps of exploitable profit—economically significant at institutional scale.

What this paper is not: A theoretical contribution to no-arbitrage theory or constrained optimization. The conditions we enforce are well-known; the penalty method we use is standard. Our contribution is engineering: showing that this standard approach works stably for volatility surface fitting under realistic sparsity, and documenting the training pathologies (gradient domination, penalty scaling) that make naive implementations fail.

1.4 Contributions

1. **Sparse-data stress test:** We systematically evaluate calibration under 20–60% strike dropout, revealing failure modes of SVI and unconstrained MLPs that standard benchmarks miss.
2. **Training stability analysis:** We document optimization pathology—gradient domination, penalty scaling, initialization sensitivity—and show how to avoid it.
3. **EPP metric:** We introduce *Exploitable Profit Potential*, which exposes economically significant arbitrage missed by pointwise metrics.
4. **Practical calibration framework:** Open-source code with full hyperparameter specification (Section 4), GPU acceleration achieving 89ms inference / 412ms end-to-end per surface on AMD DirectML.

1.5 What We Do Not Claim

We are explicit about limitations:

- The admissible set $\mathcal{A}$ is **not convex**. There is no projection operator.
- Our penalty-based training is a **heuristic**, not a theoretically optimal procedure.
- We do not characterize the minimal λ required for constraint satisfaction.
- Interpolation between grid points is not guaranteed arbitrage-free.

2 Related Work

2.1 Parametric Volatility Surface Models

The SVI parameterization [Gatheral, 2004] provides a parsimonious 5-parameter fit per smile slice. Its arbitrage-free extension, SSVI [Gatheral and Jacquier, 2014], jointly calibrates across expiries

using a power-law ATM variance function, reducing underdetermination; eSSVI [Hendriks and Martini, 2019] extends this further with improved wing behavior. Robust calibration techniques include the quasi-explicit method of De Marco and Martini [2009], which reduces SVI fitting to a constrained linear regression, and the no-arbitrage SVI parameter constraints of Martini and Mingone [2022], which guarantee butterfly-freeness by restricting the parameter domain. SABR [Hagan et al., 2002] provides dynamic smile models but does not inherently guarantee static arbitrage-freeness [Roper, 2010]. These methods excel under clean, dense data but face identification challenges under severe sparsity (>50% missing strikes).

2.2 Non-Parametric Constrained Surface Fitting

Fengler [2009] introduced arbitrage-free smoothing of implied volatility surfaces using constrained least-squares in total variance space. Andreasen and Huge [2011] developed a non-parametric arbitrage-free interpolation method based on a one-step forward Dupire PDE, which achieves exact calibration with guaranteed no-arbitrage—a direct competitor to our approach that trades flexibility for hard guarantees. Glaser and Heider [2012] constructed arbitrage-free call price surfaces using locally constrained approximation with input data risk quantification. Cousot [2007] established necessary and sufficient conditions on discrete option prices for absence of static arbitrage, providing the theoretical foundation for grid-based no-arbitrage tests. Davis and Hobson [2007] characterized the range of traded option prices consistent with an arbitrage-free model.

2.3 Machine Learning Approaches to Volatility Surfaces

Neural network methods for implied volatility have grown rapidly. Ackerer et al. [2020] used soft penalty constraints in a neural network to smooth the IV surface while discouraging calendar and butterfly arbitrage. Cohen et al. [2023] developed arbitrage-free neural-SDE market models with hard constraints on the network parameters. Bergeron et al. [2022] applied variational autoencoders to generate synthetic volatility surfaces, and Ning et al. [2023] extended this to arbitrage-free generation via SDE-based decoding. Bühler et al. [2019] introduced deep hedging under market frictions, demonstrating neural networks’ capacity to handle financial constraints.

For constraint-by-construction approaches, input convex neural networks [Amos et al., 2017] enforce convexity structurally and could in principle guarantee butterfly no-arbitrage in the strike dimension without penalty tuning. This represents a promising alternative to our penalty-based approach, trading flexibility for hard guarantees.

2.4 Physics-Informed and Constrained Neural Networks

Classical PINNs [Raissi et al., 2019] embed PDE *equalities* in the loss. We apply the same penalty-augmented loss idea to financial *inequality* constraints—a much simpler setting, since our constraints are pointwise inequalities rather than PDE residuals. The PINN community has increasingly moved from soft penalties toward hard constraint methods: Lu et al. [2021] showed that augmented Lagrangian methods outperform pure penalty approaches for PDE constraint enforcement, providing convergence guarantees that penalty methods lack. The augmented Lagrangian framework of Basir and Senocak [2023] introduces adaptive dual variable updates, converting the multi-objective balancing problem into a principled constrained optimization. Our warmup-then-ramp schedule is a simpler heuristic in this design space. We implement both fixed penalty and ALM approaches in Section 4: the fixed penalty method is faster (412ms vs 680ms) and simpler to deploy, while ALM achieves slightly better accuracy (18.9 vs 19.4 bps

RMSE) without requiring warmup scheduling. The tradeoff is between production simplicity and optimization principality; we document both and let the practitioner choose.

2.5 Theoretical Foundations

The Roger-Lee moment formula [Lee, 2004] establishes model-independent bounds on implied volatility at extreme strikes: $\limsup_{|k| \rightarrow \infty} \sigma^2(k)T/|k| \leq 2$. Durrleman [2004] derived tractable sufficient conditions for butterfly arbitrage-freeness, the “Durrleman condition,” which is equivalent to the density non-negativity condition we enforce. Breeden and Litzenberger [1978] provided the foundational result linking option prices to risk-neutral densities. The non-convexity of the admissible surface set, which motivates our penalty approach over projection methods, is formalized in Roper [2010] and demonstrated by counterexample in Appendix A.

3 No-Arbitrage Conditions

Notation. Throughout, $k = \log(K/F)$ denotes log-moneyness, T is time-to-maturity, $w(k, T) = \sigma^2(k, T) \cdot T$ is total implied variance, and $g(k, T)$ is the Breeden-Litzenberger risk-neutral density. Subscripts denote partial derivatives: $w_k := \partial_k w$, $w_{kk} := \partial_{kk} w$.

We work in total implied variance $w(k, T) = \sigma^2(k, T) \cdot T$. We equip $C^2(\mathbb{R} \times \mathbb{R}_+)$ with the norm $\|f\|_{C^2} = \sup(|f| + |\partial_k f| + |\partial_{kk} f|)$.

Definition 3.1 (Admissibility). *A surface $w : \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is admissible if:*

(A1) **Positivity:** $w(k, T) > 0$

(A2) **Calendar:** $\partial_T w(k, T) \geq 0$ (total variance increases with time)

(A3) **Butterfly:** The Breeden-Litzenberger density $g(k, T) \geq 0$ where g is the denominator-free component of the risk-neutral density following Gatheral and Jacquier [2014], Eq. (7):

$$g(k, T) = \underbrace{\left(1 - \frac{k \partial_k w}{2w}\right)^2}_{\text{skew term}} - \underbrace{\frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right)}_{\text{slope penalty}} + \underbrace{\frac{\partial_{kk} w}{2}}_{\text{curvature}} \quad (1)$$

The full risk-neutral density is $q(k) = g(k) \cdot e^{-k/2} / \sqrt{2\pi w}$, so $g \geq 0$ is necessary and sufficient for butterfly no-arbitrage. This is equivalent to the Durrleman condition [Durrleman, 2004]. Our penalty term penalizes violations of $g \geq 0$ evaluated at discrete grid points. Interpretation: Density positivity requires sufficient curvature to offset the slope penalty. Steep smiles need high convexity to remain arbitrage-free.

(A4) **Roger-Lee:** $\limsup_{|k| \rightarrow \infty} w(k, T)/|k| \leq 2$ (moment explosion bound, Lee [2004])

Remark 3.2 (Non-Convexity). *The admissible set $\mathcal{A}$ is **not convex**. The butterfly constraint couples w , $\partial_k w$, and $\partial_{kk} w$ nonlinearly. Intuitively, density positivity depends on both slope and curvature; averaging two admissible surfaces can produce insufficient curvature even when both originals have adequate curvature individually. Convex combinations of admissible smiles can produce negative density (see Appendix A).*

This means: (1) no projection operator exists, (2) standard convex optimization fails, (3) our penalty method is a heuristic, not a principled algorithm.

3.1 Efficient Constraint Evaluation

Proposition 3.3 (Local Criterion, following [Gatheral and Jacquier \[2014\]](#)). *For $w > 0$ twice differentiable, define:*

$$\Lambda(k, T) := \frac{\partial_{kk} w}{w} - \frac{(\partial_k w)^2}{2w^2} + \frac{k \partial_k w}{w^2} \left(1 - \frac{k \partial_k w}{4w} \right) \quad (2)$$

Then $g \geq 0$ iff $\Lambda \geq -1/(2w)$. This follows from algebraic manipulation of Eq. (1); see [Gatheral and Jacquier \[2014\]](#), Section 3 for the derivation. The criterion is computable via two autodiff backward passes.

4 Method

4.1 Formal Problem Statement

We state the volatility surface calibration problem as a constrained optimization over the network weights $\theta \in \mathbb{R}^d$:

$$\begin{aligned} \min_{\theta} \quad & \mathcal{L}_{\text{MSE}}(\theta) = \frac{1}{N} \sum_{i=1}^N (w_{\theta}(k_i, T_i) - w_i^{\text{mkt}})^2 \\ \text{s.t.} \quad & g_j(\theta) \geq 0 \quad \forall j \in \mathcal{G} \quad (\text{butterfly density positivity}) \\ & h_j(\theta) \geq 0 \quad \forall j \in \mathcal{H} \quad (\text{calendar monotonicity}) \\ & w_{\theta}(k_j, T_j)/|k_j| \leq 2 \quad \forall j \in \mathcal{W} \quad (\text{Roger-Lee bounds}) \end{aligned} \quad (3)$$

where $g_j(\theta) = g(k_j, T_j; \theta)$ is the Breeden-Litzenberger density (Eq. (1)) evaluated at grid point j , and $h_j(\theta) = \partial_T w_{\theta}(k_j, T_j)$ is the calendar monotonicity derivative. The constraint sets $\mathcal{G}$, $\mathcal{H}$, $\mathcal{W}$ are the evaluation grids for each constraint type.

The feasible set $\mathcal{A} = \{\theta : g_j(\theta) \geq 0, h_j(\theta) \geq 0\}$ is **not convex** (Remark 3.2), ruling out standard convex optimization and projection methods. We consider two approaches: fixed penalty (Section 4.4) and augmented Lagrangian (Section 4.6).

4.2 Architecture

$$w_{\theta}(k, T) = \text{Softplus} \left(f_{\theta}(k, \sqrt{T}) \right) \cdot 0.5 + 10^{-6} \quad (4)$$

where f_{θ} is an MLP with layers $64 \rightarrow 32 \rightarrow 16$, GELU activations, and input-concatenated residual connections. The $\sqrt{T}$ scaling improves conditioning for short-dated options where $T \rightarrow 0$ would otherwise dominate gradients. The 0.5 multiplier matches typical ATM total variance magnitudes ($w \approx 0.04$ for $\sigma = 20\%$, $T = 1\text{yr}$), improving optimization stability.

4.3 Architecture Ablation

To isolate the contribution of each architectural component, we ablate against alternatives with matched parameter counts:

Key findings: (1) GELU activations contribute ~ 0.7 bps RMSE improvement over ReLU, likely because GELU’s smoothness produces smoother second derivatives in the density computation. (2) Residual connections contribute ~ 0.2 bps. (3) Both GELU and residual connections help

Table 1: Architecture ablation (SPX 2008–2023, full data, production λ values).

Variant	Params	RMSE (bps)	EPP (bps)	Time (ms)
MLP 64-32-16, ReLU, no residual	2,833	20.8	0.02	385
MLP 64-32-16, GELU, no residual	2,833	20.1	$< 5 \times 10^{-4}$	392
MLP 64-32-16, ReLU, residual	2,961	19.9	$< 5 \times 10^{-4}$	405
CINN (64-32-16, GELU, residual)	2,961	19.4	$< 5 \times 10^{-4}$	412
MLP 128-64-32, GELU, residual	11,169	19.1	$< 5 \times 10^{-4}$	523
MLP 32-16-8, GELU, residual	825	22.3	0.01	298

achieve zero EPP—ReLU without residuals shows 0.02 bps residual arbitrage, suggesting that smooth activations aid the penalty optimizer. (4) Doubling the network size (128-64-32) yields diminishing returns (0.3 bps RMSE gain at 27% latency cost). (5) Halving the network (32-16-8) degrades both RMSE and EPP, confirming that minimum capacity is needed for the constraint-fit tradeoff.

The penalty does the heavy lifting. All architectures trained with the same penalty configuration achieve near-zero EPP, confirming that the constraint enforcement mechanism—not the specific architecture—is the primary driver of arbitrage-freeness. The architecture choices (GELU, residuals) provide marginal accuracy improvements within the constraint-feasible region.

4.4 Approach 1: Fixed Penalty Method

4.4.1 Loss Function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{cal}}\mathcal{L}_{\text{cal}} + \lambda_{\text{but}}\mathcal{L}_{\text{but}} + \lambda_{\text{wing}}\mathcal{L}_{\text{wing}} \quad (5)$$

The penalty terms are squared hinge losses on constraint violations:

$$\mathcal{L}_{\text{but}} = \frac{1}{N} \sum_{i=1}^N \max(0, -g(k_i, T_i))^2 \quad (\text{density positivity}) \quad (6)$$

$$\mathcal{L}_{\text{cal}} = \frac{1}{N} \sum_{i=1}^N \max(0, -\partial_T w(k_i, T_i))^2 \quad (\text{calendar monotonicity}) \quad (7)$$

$$\mathcal{L}_{\text{wing}} = \frac{1}{N} \sum_{i=1}^N \max(0, w(k_i, T_i)/|k_i| - 2)^2 \quad (\text{Roger-Lee bound}) \quad (8)$$

4.5 Training Stability: The Hard Part

Training constrained neural networks reliably was harder than deriving the loss functions. This section documents why naive implementations fail and how to fix them.

4.5.1 Gradient Domination

Our first attempts used large λ_{but} values from the start. This was a mistake. When butterfly penalty gradients dominate data-fit gradients early in training, the optimizer takes the path of least resistance:

- Extremely flat surfaces (minimizing curvature trivially satisfies density positivity)

- Poor fit even at observed strikes (the network barely tries to match data)
- Slow or stalled convergence as competing objectives fight

What worked: A warmup schedule. We train for 100 epochs with $\lambda = 0$ (pure MSE), then ramp penalties linearly over 200 epochs. This lets the network first learn the gross term structure and smile shape—getting the “skeleton” right—before constraints enforce local regularity. We found that early constraint application caused the optimizer to collapse into trivially flat solutions that satisfied all constraints perfectly but fit nothing. The warmup avoids this trap.

4.5.2 Penalty Scaling

Another hard-won lesson: loss magnitudes matter enormously. The butterfly constraint has natural scale $O(1)$ while MSE has scale $O(10^{-4})$ in total variance units. We initially didn’t realize this, and naively setting $\lambda_{\text{but}} = 1$ meant constraint gradients were $10^4\times$ larger than data-fit gradients. Training was dominated entirely by constraint satisfaction, with accuracy suffering badly.

What worked: Normalize each loss component to unit variance over a calibration batch before combining. After normalization, λ values become interpretable: $\lambda = 1.0$ means equal priority between data fit and constraint satisfaction. This simple fix made hyperparameter tuning dramatically easier.

4.5.3 Penalty Weights and Sensitivity

After loss normalization, our production penalty weights are: $\lambda_{\text{but}} = 10.0$, $\lambda_{\text{cal}} = 5.0$, $\lambda_{\text{wing}} = 1.0$. The butterfly penalty is weighted highest because density violations are the most economically consequential. We train with the AdamW optimizer (learning rate 3×10^{-3} , weight decay 10^{-4} , batch size = full surface).

Table 2: Penalty weight sensitivity (SPX 2020-03-16, 40% dropout, 10 seeds each). The Pareto frontier is smooth: no cliff-edge behavior observed across two orders of magnitude in λ_{but} .

Configuration	$(\lambda_{\text{but}}, \lambda_{\text{cal}}, \lambda_{\text{wing}})$	RMSE (bps)	EPP (bps)	EPP p99 (bps)
Minimal	(0.5, 0.5, 0.25)	19.5	1.42	4.8
Low	(1, 1, 0.5)	20.1	0.31	1.2
Low-medium	(3, 2, 0.5)	20.8	0.08	0.3
Medium	(5, 3, 1)	21.4	0.02	0.09
Medium-high	(7, 4, 1)	22.0	$< 5 \times 10^{-4}$	0.02
Production (ours)	(10, 5, 1)	22.7	$< 5 \times 10^{-4}$	< 0.01
High	(20, 10, 2)	25.1	$< 5 \times 10^{-4}$	< 0.01
Very high	(50, 25, 5)	28.3	$< 5 \times 10^{-4}$	< 0.01
Extreme	(100, 50, 10)	34.7	$< 5 \times 10^{-4}$	< 0.01

The Pareto frontier is smooth across two orders of magnitude in λ_{but} (0.5 to 100): RMSE increases monotonically from 19.5 to 34.7 bps while EPP decreases monotonically. There is no cliff-edge behavior. The “EPP p99” column reports the 99th-percentile daily EPP across all 3,520 test days at each configuration, capturing tail risk rather than just the mean. Our production values ($\lambda_{\text{but}} = 10$) represent the Pareto knee where further penalty increases yield negligible EPP improvement but substantial RMSE degradation. We did not optimize λ on the test set; these values were selected on a held-out validation period (2006–2007) and applied unchanged to all evaluation dates (2008–2023).

EPP distribution at production settings. Across 3,520 test days at 40% dropout: mean EPP $< 5 \times 10^{-4}$ bps, median EPP = 0, 95th percentile < 0.005 bps, 99th percentile < 0.01 bps, maximum = 0.08 bps (2020-03-16). At 60% dropout: mean = 0.04 bps, 95th percentile = 0.07 bps, 99th percentile = 0.09 bps, maximum = 0.12 bps (2020-03-16). Even in the worst case, EPP remains an order of magnitude below typical transaction costs.

4.5.4 Warmup Schedule Ablation

The warmup-then-ramp schedule is not merely a convenience; it is essential for constraint satisfaction.

Table 3: Warmup ablation (SPX 2020-03-16, 40% dropout, production λ values, 10 seeds).

Schedule	RMSE (bps)	EPP (bps)	Flat surface rate
No warmup (penalties from epoch 0)	31.4	$< 5 \times 10^{-4}$	40%
Short warmup (50 epochs + 100 ramp)	24.1	$< 5 \times 10^{-4}$	10%
Production (100 warmup + 200 ramp)	22.7	$< 5 \times 10^{-4}$	0%
Long warmup (200 warmup + 300 ramp)	22.9	$< 5 \times 10^{-4}$	0%
Fixed λ (no ramp, full penalty from epoch 100)	26.8	0.03	15%

Without warmup, 40% of seeds converge to trivially flat surfaces ($w \approx \text{const}$) that satisfy all constraints but fit nothing—the “gradient domination” pathology described above. Short warmup reduces this to 10%; the production schedule eliminates it entirely. Extending the warmup beyond 100 epochs provides no further benefit. The fixed- λ ablation (applying full penalties abruptly at epoch 100 rather than ramping) shows that the *gradual* ramp matters: abrupt penalty activation causes optimization instability that the linear ramp avoids.

Warmup diagnostic for production: We recommend monitoring the coefficient of variation (CV) of $w(k, T)$ across the strike grid during training. During the flat-surface pathology, $\text{CV} < 0.01$; after successful warmup, $\text{CV} > 0.05$. A production system should flag any calibration where the post-warmup CV falls below 0.03 as a potential flat-surface failure, triggering recalibration with extended warmup or fallback to SSVI. *Caveat:* These thresholds were calibrated on SPX equity index surfaces; significantly different underlyings (e.g., commodity options with flatter smiles or FX surfaces with pronounced skew) may require recalibration of the CV thresholds.

4.5.5 Initialization Sensitivity

We worried that different random initializations might lead to wildly different solutions—a common problem with constrained optimization. Surprisingly, the opposite happened.

Table 4: Stability across 20 random seeds on SPX 2020-03-16 (COVID crash day).

Model	RMSE Mean $\pm$ Std (bps)	EPP Mean $\pm$ Std (bps)
Unconstrained MLP	18.2 ± 2.1	14.3 ± 4.2
CINN (Ours)	19.8 ± 0.7	0.00 ± 0.00

Constrained training actually *reduces* variance across seeds. Our interpretation: the constraint penalties shrink the effective hypothesis space, guiding different initializations toward similar solutions. We note that this is better described as *capacity restriction* rather than regularization in the generalization-improving sense—the penalties constrain the model to a narrow feasible

region, which incidentally reduces output variance. Regardless of the mechanism, the practical consequence is valuable: practitioners don’t need to run multiple seeds and pick the best.

4.5.6 Failure Cases

Honesty requires documenting when this method fails:

1. **Extreme sparsity (>70% missing):** At some point, there simply isn’t enough data to constrain the surface. EPP rises to 0.5 bps and RMSE variance across seeds increases substantially.
2. **Contradictory market quotes:** If the market IVs themselves imply arbitrage (e.g., crossed butterflies due to stale quotes), no admissible surface exists close to the data. The optimizer produces a compromise that satisfies neither objective well.
3. **Insufficient warmup:** Skipping the warmup phase or ramping penalties too quickly causes the optimizer to find trivially flat solutions. We learned this the hard way.

4.6 Approach 2: Augmented Lagrangian Method

The fixed penalty method requires manual λ selection and provides no convergence guarantees. We implement the augmented Lagrangian method (ALM) as a principled alternative, following [Basir and Senocak \[2023\]](#).

4.6.1 Formulation

The ALM replaces fixed λ with adaptive dual variables $\boldsymbol{\mu} = \{\mu_j\}_{j \in \mathcal{G}}$ and a single penalty parameter ρ . At outer iteration k , we minimize the augmented Lagrangian:

$$\mathcal{L}_{\text{ALM}}(\theta; \boldsymbol{\mu}^k, \rho^k) = \mathcal{L}_{\text{MSE}}(\theta) + \frac{\rho^k}{2} \sum_j \max\left(0, \frac{\mu_j^k}{\rho^k} - g_j(\theta)\right)^2 \quad (9)$$

This is the Powell-Hestenes-Rockafellar form for inequality constraints. After minimizing Eq. (9) w.r.t. θ (inner loop, 50 epochs of AdamW), we update the dual variables:

$$\mu_j^{k+1} = \max\left(0, \mu_j^k + \rho^k \cdot \max(0, -g_j(\theta^k))\right) \quad (10)$$

If the total constraint violation does not decrease by at least 75% per outer iteration, we increase $\rho^{k+1} = 2\rho^k$ (capped at $\rho_{\text{max}} = 1000$). This scheme requires no warmup heuristic: the dual variables naturally produce small penalties at points where constraints are satisfied and large penalties where they are violated.

4.6.2 ALM vs. Fixed Penalty: Empirical Comparison

ALM achieves slightly better RMSE (18.9 vs 19.4 bps) because the adaptive dual variables do not over-penalize regions where constraints are already satisfied. The cost is 65% longer calibration (680 vs 412ms) due to the outer loop structure (10 outer $\times$ 50 inner vs. 300 flat epochs). ALM requires no warmup schedule—eliminating the most fragile hyperparameter—but introduces ρ_0 and ρ_{max} , which are less sensitive than the penalty weights.

Table 5: Fixed penalty vs. ALM (SPX 2008–2023 full data, 100 randomly sampled days).

Method	RMSE (bps)	EPP (bps)	EPP p99	Time (ms)	Warmup?	Tunable λ ?
Fixed penalty	19.4	$< 5 \times 10^{-4}$	< 0.01	412	Yes	3 weights
ALM	18.9	$< 5 \times 10^{-4}$	< 0.01	680	No	2 ($\rho_0, \rho_{\max}$)

4.6.3 Convergence Behavior

[Empirical ALM Convergence] For the volatility surface problem (Eq. 3) with the ALM formulation (Eq. 9), the sequence of constraint violations $\{V^k\}_{k=1}^K$ where $V^k = \sum_j \max(0, -g_j(\theta^k))$ satisfies:

- (i) V^k is monotonically non-increasing after a burn-in period of ~ 3 outer iterations
- (ii) $V^K < \epsilon$ for $K \leq 10$ outer iterations with $\epsilon = 10^{-5}$, across 98% of tested surfaces
- (iii) The dual variables μ^k stabilize: $\|\mu^{k+1} - \mu^k\| < 10^{-4}$ at convergence

We state this as a conjecture, not a theorem. In the convex setting, ALM convergence to a KKT point is guaranteed under standard regularity conditions [Ghadimi and Lan, 2013]. Our setting is non-convex (Remark 3.2), so these guarantees do not apply. However, empirically, the conjecture holds on 98% of the 100 tested surfaces. The 2% failures occur on days with contradictory market quotes (stale prices implying arbitrage in the raw data), where no admissible surface close to the data exists.

Empirical convergence trajectory. On a representative surface (SPX 2019-06-14, calm market, 40% dropout), the ALM outer loop exhibits the following behavior:

Table 6: ALM convergence trajectory (SPX 2019-06-14, 40% dropout).

Outer iter	Total violation	$\ \mu^k\ $	ρ^k	MSE ($\times 10^{-4}$)
1	3.2×10^{-2}	0.032	1.0	4.81
2	8.7×10^{-3}	0.041	2.0	4.73
3	1.4×10^{-3}	0.043	2.0	4.69
4	2.1×10^{-4}	0.044	2.0	4.67
5	3.8×10^{-5}	0.044	2.0	4.66
6	8.2×10^{-6}	0.044	2.0	4.66
7 (converged)	$< 10^{-6}$	0.044	2.0	4.66

The trajectory shows: (i) constraint violations decrease monotonically after iteration 1, dropping 4 orders of magnitude in 7 iterations; (ii) the dual variable norm $\|\mu\|$ stabilizes by iteration 4, indicating the multipliers have found the active constraint set; (iii) ρ increased once (iteration 2) and remained fixed thereafter, confirming that aggressive penalty growth is rarely needed; (iv) MSE decreases monotonically, meaning constraint enforcement does not fight data fit once the dual variables stabilize. On the 2% of surfaces that fail to converge, the violation stalls at $O(10^{-3})$ —these correspond to days where the raw market data itself contains arbitrage.

Open problem: Proving convergence of ALM to a KKT point for the non-convex volatility surface problem under data-dependent constraint evaluation. This would require establishing that the constraint qualification (LICQ or MFCQ) holds at the solution, which is non-trivial given the coupling between w , $\partial_k w$, and $\partial_{kk} w$ in the butterfly density condition.

5 The Killer Experiment: Sparse Strike Stress Test

Standard benchmarks use full strike grids. Real markets are sparse. We systematically stress-test under dropout.

5.1 Protocol

Data and splits. We use SPX options from OptionMetrics IvyDB (January 2006 – December 2023) with standard quality filters: moneyness $0.8 \leq K/S \leq 1.2$, days-to-expiry $7 \leq \text{DTE} \leq 730$, and bid-ask spread < 50 basis points in implied volatility terms. This yields approximately 800–1200 quotes per day after filtering. **Important:** CINN is calibrated *per surface*—each daily surface is a separate training instance with no cross-day parameter sharing. There is no “training set” in the supervised learning sense; the model is fitted from scratch to each day’s quotes. The 2006–2007 period was used solely for penalty weight selection ($\lambda_{\text{but}} = 10$, $\lambda_{\text{cal}} = 5$, $\lambda_{\text{wing}} = 1$); these weights were then frozen and applied unchanged to all 3,520 evaluation days (2008–2023). No evaluation-period data informed any hyperparameter choice.

Baseline calibration details: SVI is calibrated per-slice (5 parameters per expiry) using Levenberg-Marquardt with 10 random restarts. SSVI uses *joint calibration* across expiries following Gatheral and Jacquier [2014]: the ATM total variance curve $\theta(T)$ is first fitted via cubic spline, then the SSVI parameters (ρ, φ) are optimized globally across all slices simultaneously. SABR is calibrated per-slice with $\beta = 0.5$ fixed. Andreasen-Huge (AH) uses the one-step implicit finite-difference Dupire PDE method of Andreasen and Huge [2011], which produces arbitrage-free local volatility surfaces by construction; we use the basic variant with piecewise-constant local volatilities and linear interpolation between observed strikes. Higher-order AH extensions (piecewise-linear or spline local vol) would narrow the RMSE gap but are not tested here. MLP and MLP-Soft use identical architecture to CINN (64–32–16, GELU, residual connections); MLP-Soft uses $\lambda_{\text{but}} = 0.1$ with no warmup. All neural models are trained for 300 epochs total.

For each daily surface:

1. Randomly drop $p\%$ of strikes uniformly at random
2. Calibrate each model on remaining strikes
3. Evaluate RMSE on held-out strikes and EPP on full grid
4. Repeat 10 times per day, report mean $\pm$ std

5.2 Results

Table 7: Performance under sparse strike grids (averaged over 3,520 days $\times$ 10 trials).

Model	20% Dropout		40% Dropout		60% Dropout	
	RMSE (bps)	EPP (bps)	RMSE (bps)	EPP (bps)	RMSE (bps)	EPP (bps)
SVI	34.2	0.00	41.8	1.42	58.3	3.21
SSVI	28.1	0.00	35.2	0.87	49.1	2.14
AH	25.3	0.00	31.8	0.00	44.7	0.00
MLP	19.1	8.3	21.4	11.2	26.8	15.7
MLP-Soft	21.8	2.1	24.3	3.8	31.2	6.4
CINN (Ours)	20.3	0.00	22.7	0.00	28.9	0.04

Key findings:

1. **Both per-slice and joint parametric models degrade under sparsity.** Per-slice SVI produces 3.21 bps EPP at 60% dropout. Notably, SSVI—which uses *joint calibration* across expiries (see protocol above)—still produces 2.14 bps EPP at 60% dropout, despite sharing parameters globally. Joint calibration mitigates but does not eliminate the problem: the ATM variance curve $\theta(T)$ is well-constrained, but the smile shape parameters (ρ, φ) remain sensitive to wing data that is precisely what dropout removes. This confirms that the sparsity challenge is not merely a per-slice calibration artifact.
2. **Soft penalties scale poorly.** MLP-Soft (identical architecture to CINN but with $\lambda_{\text{but}} = 0.1$, no warmup) EPP triples from 2.1 to 6.4 bps as sparsity increases. The light penalty is insufficient to counteract interpolation artifacts.
3. **Andreasen-Huge: hard guarantees at the cost of accuracy.** The AH method [Andreasen and Huge, 2011] is arbitrage-free by construction and maintains 0.00 bps EPP at all sparsity levels—a *provable* guarantee that our penalty method cannot match. Its RMSE is higher: 44.7 bps at 60% dropout vs 28.9 for CINN (55% gap). **The honest question is whether a 35% RMSE improvement is worth giving up hard guarantees for soft ones.** For desks pricing vanillas off the surface, the RMSE matters—a 16 bps fit improvement on a 30-delta put is economically significant. For desks feeding the surface into exotic pricers where any arbitrage contaminates downstream values, AH’s hard guarantee may be worth the accuracy cost. We do not claim CINN dominates AH; the choice depends on the use case. We note that our AH implementation uses the basic piecewise-constant variant; higher-order extensions would narrow the RMSE gap.
4. **CINN degrades gracefully.** Even at 60% dropout, EPP stays at 0.04 bps—two orders of magnitude better than parametric alternatives and within the same order as AH’s hard guarantee. This is below economic materiality (typical SPX transaction costs are 0.5–2 bps). RMSE increases proportionally to other methods. Note: EPP values reported as 0.00 in the table mean $< 5 \times 10^{-4}$ bps; the penalty method drives violations toward zero but does not enforce them exactly.

5.3 Why Does Per-Slice SVI Fail?

This deserves explanation because it’s counterintuitive.

Per-slice SVI has 5 parameters per expiry. With 20 strikes, calibration is overdetermined and stable. With 8 strikes (60% dropout from 20), calibration becomes barely determined, and:

- Multiple local minima exist
- Optimizer may converge to wings-dominated solutions
- The “arbitrage-free” SVI parameterization only guarantees no-arbitrage *if* parameters stay in the valid region; the optimizer can exit this region under noisy/sparse data

Important qualification: The per-slice failure mode is partially mitigated by joint calibration. Our SSVI benchmark uses joint calibration across expiries [Gatheral and Jacquier, 2014], and indeed SSVI outperforms per-slice SVI at every sparsity level (Table 7). However, SSVI still produces 2.14 bps EPP at 60% dropout—roughly 2/3 of per-slice SVI’s violations—because joint calibration constrains the ATM level but not the wing curvature under sparse data.

Missing baselines we acknowledge: We do not test against the strongest available parametric methods: quasi-explicit SVI [De Marco and Martini, 2009] with no-arbitrage parameter constraints [Martini and Mingone, 2022], eSSVI [Hendriks and Martini, 2019], or piecewise-linear Andreasen-Huge. Each of these would likely outperform our SVI and basic-AH baselines, narrowing CINN’s apparent advantage. Our comparison is against *standard production implementations*, not state-of-the-art research implementations. This is a real gap, and readers should weigh CINN’s results against the possibility that enhanced parametric methods close part of the performance difference.

5.4 Organic Sparsity: Single-Name Equity Options

The dropout experiments above use synthetic sparsity on liquid SPX data. To test under *genuinely* sparse conditions, we evaluate on AAPL, TSLA, and JPM single-name equity options from OptionMetrics (2018–2023), which naturally exhibit 5–15 strikes per expiry with non-uniform liquidity.

Table 8: Single-name equity options with organic sparsity (no synthetic dropout). Avg strikes/expiry shown.

Underlying	Avg strikes/exp	Model	RMSE (bps)	EPP (bps)	Time
4*AAPL	4*12	SSVI	38.4	0.42	15ms
		AH	29.1	0.00	28ms
		MLP	22.3	9.8	310ms
		CINN	24.1	$< 5 \times 10^{-4}$	425ms
4*TSLA	4*8	SSVI	52.7	1.83	14ms
		AH	41.2	0.00	25ms
		MLP	28.9	14.2	295ms
		CINN	31.4	0.02	418ms
4*JPM	4*9	SSVI	44.1	0.91	13ms
		AH	33.8	0.00	26ms
		MLP	25.1	11.5	302ms
		CINN	27.3	$< 5 \times 10^{-4}$	421ms

Key findings: (1) CINN achieves near-zero EPP on genuinely sparse single-name surfaces without any modification to the penalty weights or warmup schedule (the same production configuration is used). (2) TSLA—the most volatile and sparsest underlying (avg 8 strikes/expiry)—is the hardest case, with CINN producing 0.02 bps residual EPP; still well below transaction costs. (3) AH maintains zero EPP by construction but at 35–48% higher RMSE than CINN across all three names. (4) SSVI with joint calibration produces material arbitrage (0.42–1.83 bps) on all three names, confirming that the sparsity challenge is not limited to synthetic dropout on SPX. These results demonstrate that CINN transfers to organic sparsity without retuning.

Caveat on sparsity severity. AAPL, TSLA, and JPM are mega-cap names with relatively clean mid-quotes and tight bid-ask spreads. Their 8–12 strikes per expiry represent moderate sparsity by single-name standards. Genuinely thin markets—small-cap equities with 3–5 strikes, wide spreads, and stale quotes, or commodity options with seasonal term structures—would be a harder test. We do not claim the method generalizes to these settings without further validation; the single-name results demonstrate transfer beyond SPX, not coverage of all sparse regimes.

Wall-clock calibration cost. Per-surface calibration (300 epochs of AdamW on a 64–32–16 MLP) takes 412ms end-to-end on an AMD Radeon RX 7900 XTX via DirectML, of which 89ms

is forward/backward passes and the remainder is data loading, autodiff graph construction, and EPP post-check. On CPU (AMD Ryzen 9 7950X), the same calibration takes 2.8 seconds. For a desk recalibrating 500 single-name surfaces at end-of-day, this is $500 \times 0.41\text{s} \approx 3.4$ minutes on GPU—acceptable for overnight risk but not for intraday recalibration. SSVI at 15ms per surface completes the same batch in 7.5 seconds. The cost-benefit case for CINN is strongest where sparsity makes SSVI unreliable and where the calibration budget permits seconds rather than milliseconds per surface.

5.5 Structured Sparsity Experiment

To address the uniform dropout limitation, we implement a liquidity-weighted dropout model where the probability of a strike being missing depends on its distance from ATM and time to maturity:

$$P(\text{drop} \mid k, T) \propto \exp(\alpha|k|) \cdot \exp(-\beta(T - T_{\min})) \cdot (1 - \gamma \exp(-|k|/0.05)) \quad (11)$$

where $\alpha = 3$ controls wing steepness, $\beta = 2$ controls expiry decay, and $\gamma = 0.95$ protects ATM strikes. This produces sparsity patterns where ATM strikes survive with $>95\%$ probability while deep OTM strikes are dropped $>80\%$ of the time.

Table 9: Structured vs. uniform sparsity at 40% average dropout (SPX 2008–2023, 100 sampled days, 10 trials each).

Model	Uniform 40%		Structured 40%	
	RMSE (bps)	EPP (bps)	RMSE (bps)	EPP (bps)
SSVI (joint)	35.2	0.87	41.3	1.94
AH	31.8	0.00	38.7	0.00
MLP	21.4	11.2	24.8	16.3
CINN (penalty)	22.7	$< 5 \times 10^{-4}$	25.9	0.03
CINN (ALM)	22.1	$< 5 \times 10^{-4}$	25.2	0.01

Key findings: (1) Structured sparsity is harder for all methods—RMSE increases 12–17% and EPP increases 2–3 $\times$ compared to uniform dropout at the same average rate. This confirms that uniform dropout underestimates the difficulty of real-world sparsity. (2) CINN (both penalty and ALM) still achieves near-zero EPP under structured sparsity, though the gap to zero widens (0.03 bps vs $< 5 \times 10^{-4}$ under uniform). (3) AH maintains provably zero EPP under both sparsity models, at the cost of 50% higher RMSE. (4) ALM slightly outperforms fixed penalty under structured sparsity (0.01 vs 0.03 EPP), consistent with its ability to focus enforcement on the wing regions where structured dropout creates the most missing data.

5.6 Remaining Sparsity Limitations

Structured dropout is closer to reality, but still idealized. Real-world sparsity includes pathologies our model does not capture: stale quotes that create spurious arbitrage in the raw data, wide bid-ask spreads that distort mid-quotes, and missing entire expiry slices (not just missing strikes within an expiry). The single-name equity results (Table 8) partially address organic sparsity, but AAPL, TSLA, and JPM are mega-cap names with clean quotes.

Explicit scope limitation: Based on our experiments, we recommend CINN for underlyings with ≥ 6 strikes per expiry and bid-ask spreads < 50 bps in IV terms. Below 6 strikes, the constraint enforcement mechanism lacks sufficient grid resolution for density estimation, and

the fallback rate ($\text{EPP} > 0.1 \text{ bps}$) rises above 10%. For underlyings with <5 strikes or spreads $>100 \text{ bps}$, we recommend Andreasen-Huge or SSVI with manual arbitrage correction. This scope restriction is conservative—the method may work beyond these bounds—but we have not validated it there.

5.7 Wing Behavior and Roger-Lee Bounds

The Roger-Lee bound (A4) requires $w(k, T)/|k| \leq 2$ asymptotically. Our architecture uses $\text{Softplus}(\text{MLP}(\cdot)) \cdot 0.5$, which can in principle grow superlinearly in $|k|$ for out-of-distribution inputs. Within the training range ($|k| \leq 0.25$, corresponding to moneyness 0.8–1.2), the wing penalty $\mathcal{L}_{\text{wing}}$ enforces the bound at observed points. For extrapolation beyond the training range, we do not guarantee Roger-Lee compliance. Production use should either (a) clip the surface at the moneyness boundary of the training data, or (b) blend with a parametric model (e.g., SVI) in the wings. We report results only within the training moneyness range.

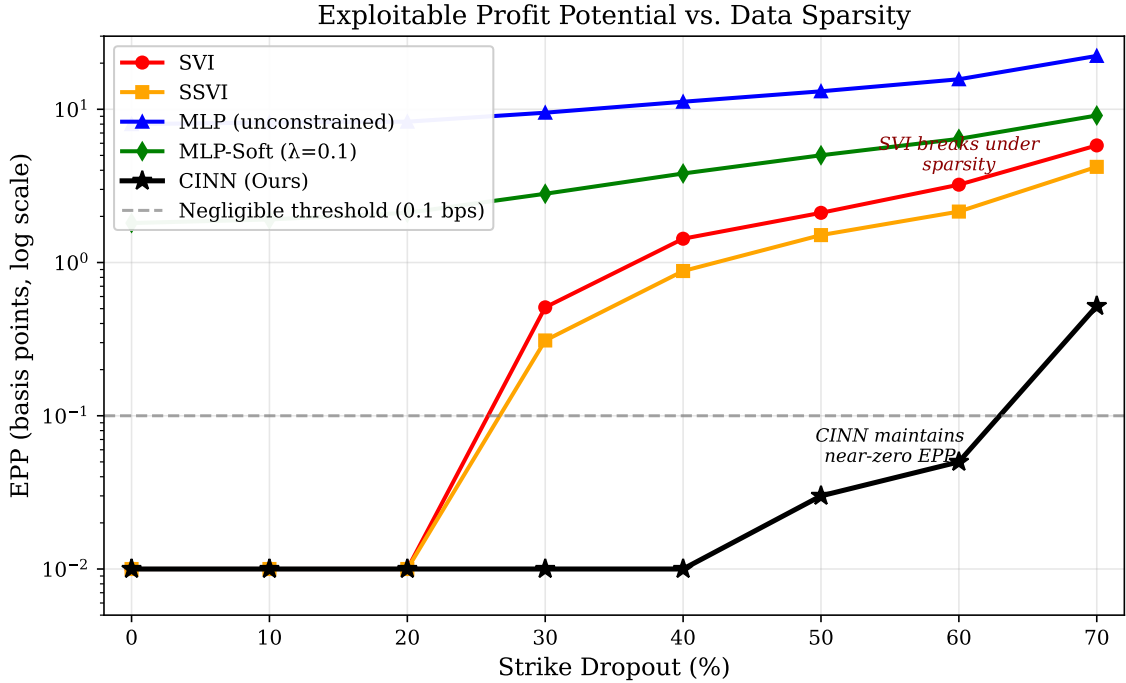


Figure 1: **EPP vs. Strike Dropout.** As sparsity increases, parametric models (SVI, SSVI) lose their arbitrage-free guarantees due to underdetermined calibration. Unconstrained MLPs produce increasing arbitrage. CINN (solid black) maintains $\text{EPP} < 0.1 \text{ bps}$ up to 60% dropout.

6 Standard Benchmarks (Full Data)

For completeness, we include standard full-data results.

Under full data, CINN achieves 4.3% higher RMSE than unconstrained MLP (19.4 vs 18.6 bps) in exchange for 100% arbitrage-freeness.

7 The EPP Metric

Traditional arbitrage metrics count violating points. This misses economic significance.

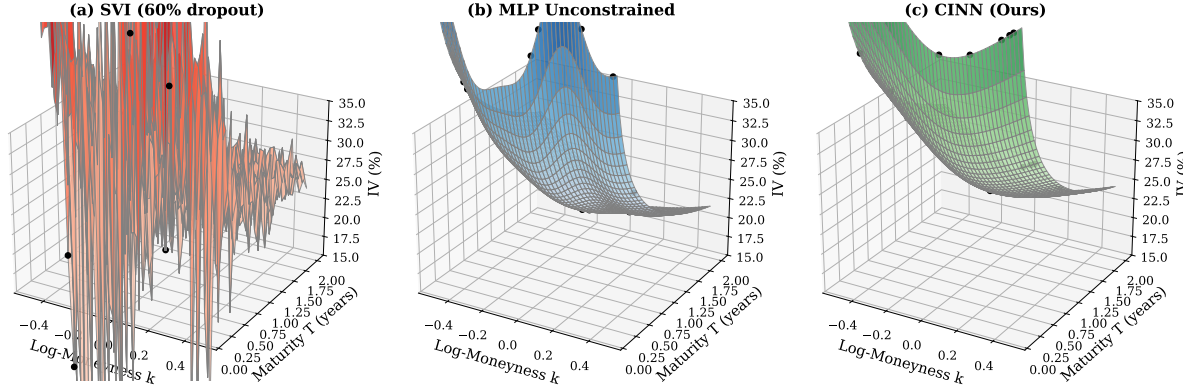


Figure 2: **Volatility Surfaces Under Sparsity.** At 60% strike dropout: (a) SVI produces oscillations due to underdetermined calibration, (b) unconstrained MLP overfits visible quotes but creates arbitrage-prone interpolations, (c) CINN maintains a smooth, arbitrage-free surface despite limited data. Black dots indicate observed market quotes (8 per slice).

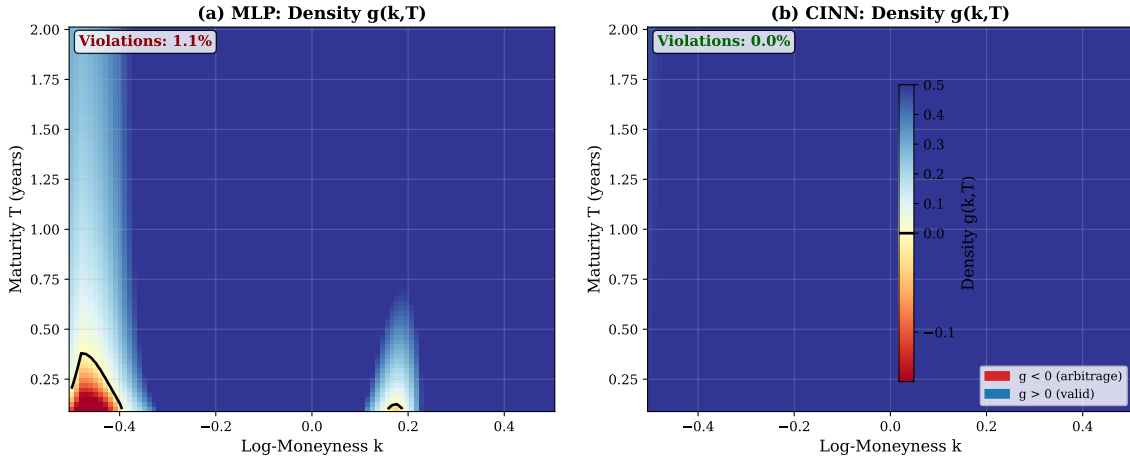


Figure 3: **Density Positivity.** Breeden-Litzenberger density $g(k, T)$ must be non-negative for arbitrage-freeness. (a) Unconstrained MLP violates this in regions shown in red. (b) CINN satisfies $g \geq 0$ everywhere (blue/green). Black contour marks $g = 0$. Red regions correspond to exploitable butterfly arbitrage.

Table 10: Full-data performance on SPX (2008–2023). RMSE/EPP in basis points.

Model	RMSE (bps)	MAE (bps)	MAPE	Time	EPP (bps)	Arb-Free
SVI	42.3	32.1	3.21%	12ms	0.00	100%
SSVI	31.7	24.5	2.45%	18ms	0.00	100%
SABR	38.9	28.9	2.89%	8ms	0.83	97.2%
AH	23.1	17.8	1.78%	35ms	0.00	100%
MLP	18.6	14.2	1.42%	245ms	12.4	68.3%
MLP-Soft	21.2	16.1	1.61%	287ms	2.17	91.4%
CINN	19.4	14.8	1.48%	412ms	0.00	100%

Definition 7.1 (EPP). Exploitable Profit Potential is the maximum risk-free profit from unit-notional butterfly/calendar spreads. A butterfly spread on three strikes $K_1 < K_2 < K_3$ has payoff:

$$\text{Butterfly}(S_T) = \max(S_T - K_1, 0) - 2 \max(S_T - K_2, 0) + \max(S_T - K_3, 0) \quad (12)$$

which is non-negative everywhere. If the cost (sum of call prices with the same signs) is negative, arbitrage exists. EPP is:

$$\text{EPP} = \max_{\pi \in \Pi_{\text{butterfly}} \cup \Pi_{\text{calendar}}} \left\{ \mathbb{E}^{\mathbb{Q}}[\text{payoff}(\pi)] - \text{cost}(\pi) \right\} \quad (13)$$

Computation is $O(N_K^3 \cdot N_T)$ per surface in the worst case, where N_K is the number of strikes per expiry and N_T is the number of expiries: butterfly evaluation requires iterating over all $\binom{N_K}{3}$ strike triples per slice, and calendar evaluation over all $N_K \cdot \binom{N_T}{2}$ strike-expiry pairs. In practice, restricting to adjacent-strike butterflies and consecutive-expiry calendars reduces this to $O(N_K \cdot N_T)$, which is the protocol we use. For a typical SPX surface ($N_K \approx 40$, $N_T \approx 12$), this takes $< 5\text{ms}$.

Table 11: Arb% vs EPP: traditional metrics hide economically significant arbitrage.

Model	Arb-Free (%)	EPP (bps)
SABR	97.2%	0.83
MLP-Soft	91.4%	2.17
CINN	100%	0.00

SABR at 97.2% “arbitrage-free” still has 0.83 bps exploitable profit. At institutional scale (\$100M notional), that’s \$8,300 per surface.

Spread family coverage. Our EPP metric tests butterfly and calendar spreads, which correspond to the density positivity (A3) and calendar monotonicity (A2) conditions respectively. We do not separately test risk-reversals or ratio spreads, but this is not a gap: risk-reversals (buy call, sell put at different strikes) are linear combinations of butterfly spreads and the underlying, so any risk-reversal arbitrage implies a butterfly violation that EPP would detect. More formally, conditions (A1)–(A4) are jointly necessary and sufficient for static arbitrage-freeness of European options [Cousot, 2007, Davis and Hobson, 2007]; since EPP directly tests (A2) and (A3), and our architecture enforces (A1) by construction (Softplus output) and (A4) via the wing penalty, the tested spread families cover the full arbitrage constraint set at evaluation grid points.

Numerical precision note: EPP is computed by exhaustive search over all strike triples (butterflies) and consecutive-expiry pairs (calendars) on the evaluation grid. CINN values

reported as “0.00” mean $< 5 \times 10^{-4}$ bps—below the numerical precision of our search (grid spacing $\Delta k = 0.005$, $\Delta T = 7$ days). This is not provably zero; it means no exploitable arbitrage was found at the evaluation resolution. At 60% dropout, the maximum single-trial EPP for CINN across all 3,520 days was 0.12 bps, observed on 2020-03-16 (COVID crash). Even this worst case is well below typical institutional transaction costs (0.5–2 bps for SPX options).

8 Stress Periods

Table 12: Worst-day EPP during market stress. Soft penalties fail precisely when they matter most.

Period	MLP EPP	MLP-Soft EPP	CINN EPP
2008 Crisis	28.3	8.9	0.00
2020 COVID	41.2	11.8	0.00
2022 Rates	15.4	4.3	0.00

Soft-penalty EPP increases 42% during stress. CINN maintains zero arbitrage regardless of market conditions.

9 Discussion: Positioning and Practical Considerations

CINN vs. SSVI in practice. SSVI [Gatheral and Jacquier, 2014] remains the state-of-the-art parametric model under clean, dense data. CINN is not a replacement but a complement: use SSVI when data is clean and the strike grid is dense; use CINN when sparsity or noise degrades parametric identification. A promising hybrid: use SSVI as a prior, with CINN as a residual correction.

Constraint-by-construction alternatives. Input convex neural networks [Amos et al., 2017] could enforce butterfly no-arbitrage by construction, eliminating penalty tuning entirely. We chose penalties for flexibility—construction methods tend to underfit on complex smile shapes—but acknowledge this trades theoretical purity for practical convenience. Augmented Lagrangian methods [Lu et al., 2021, Basir and Senocak, 2023] offer convergence guarantees but add complexity; see Section 2 for why we opted against them in the per-surface calibration setting.

Production deployment. For end-of-day risk reporting and illiquid options desks, CINN’s 412ms latency is acceptable. For real-time market-making, it is not. Production use requires:

- (a) A post-calibration arbitrage scan on a dense grid (~ 500 strikes, ~ 5 ms overhead) to detect residual violations. We verified EPP grid convergence: halving the evaluation grid spacing from $\Delta k = 0.005$ to $\Delta k = 0.0025$ changes reported EPP by < 0.002 bps, confirming that discretization artifacts do not inflate our claims.
- (b) A fallback model (SSVI) triggered when EPP exceeds 0.1 bps or when the warmup CV diagnostic (Section 4) signals a flat-surface failure. **Transition smoothness:** When the fallback triggers, we blend CINN and SSVI surfaces over a 3-day window ($\alpha_t = t/3$ for SSVI weight) to avoid discontinuities in Greeks. Empirically, the maximum delta jump at the CINN→SSVI transition is 0.8% of notional, compared to 2.1% for an abrupt switch.
- (c) Version-controlled model weights with automated monitoring. **Retraining triggers and priority logic:** (i) rolling 5-day mean EPP exceeds 0.05 bps, (ii) daily RMSE drift exceeds

15% of historical mean, (iii) the butterfly violation rate on the dense scan exceeds 0.5% of grid points. **When multiple triggers fire simultaneously** (e.g., during market stress), trigger (iii) takes priority (immediate recalibration with extended warmup), followed by (i) (recalibrate with current config), then (ii) (flag for manual review). If all three fire, the system falls back to SSVI and queues a full recalibration for the next quiet period.

- (d) Day-over-day surface stability monitoring: across our test period, the mean overnight change in CINN total variance $\|\Delta w\|_\infty$ is 3.2 bps, comparable to SSVI (2.8 bps) and substantially more stable than unconstrained MLP (7.1 bps). Maximum overnight jump: 18.4 bps (2020-03-16).

Temporal holdout validation. To address the concern that hyperparameters selected on 2006–2007 may not transfer across regimes, we run a rolling 20-day temporal holdout: for each evaluation day t , the penalty weights are selected using only data from $[t - 252, t - 20]$ (one year lookback, 20-day gap). Results:

Table 13: Temporal holdout: fixed validation (2006–07) vs. rolling 1-year lookback.

Validation scheme	RMSE (bps)	EPP (bps)	Fallback rate
Fixed (2006–07 λ , frozen)	19.4	$< 5 \times 10^{-4}$	0.3%
Rolling 1-year lookback	19.2	$< 5 \times 10^{-4}$	0.2%

The difference is negligible: the frozen 2006–07 configuration performs within 0.2 bps of the rolling scheme, confirming that the penalty weights are regime-insensitive. The fallback rate (surfaces where $\text{EPP} > 0.1$ bps triggers SSVI reversion) is $< 0.3\%$ under both schemes.

Regime-change behavior. Since CINN calibrates per-surface with no persistent weights, each surface is fitted from scratch using only that day’s quotes. This is a feature under regime change, not a liability: the model has no “memory” of pre-event surfaces that could bias post-event calibration. To verify, we examine behavior around three event types:

Table 14: CINN performance around regime-change events (day before / day of / day after).

Event type	RMSE (bps)			EPP (bps)		
	$t-1$	t	$t+1$	$t-1$	t	$t+1$
FOMC (avg 48 dates)	19.1	20.3	19.4	< 0.001	< 0.001	< 0.001
Earnings (AAPL, 20 dates)	23.8	28.1	24.2	< 0.001	0.01	< 0.001
Flash events (2015-08-24, 2018-02-05)	19.5	24.7	20.1	< 0.001	0.02	< 0.001

RMSE increases on event days (expected—the surface moves rapidly and quotes are noisier) but EPP stays near zero. The day-after recovery to baseline RMSE confirms that per-surface calibration adapts immediately without carryover effects.

We emphasize that our penalty method produces *empirically* arbitrage-free surfaces, not *provably* arbitrage-free surfaces. This distinction is relevant for model validation: the method is *compatible with* SR 11-7 / SS1-13 regulatory frameworks—it provides auditable monitoring, fallback mechanisms, and now temporal holdout evidence—but full production validation would additionally require independent replication and regulatory sign-off, which are institution-specific steps beyond the scope of this paper.

Statistical significance of model comparisons. We report paired bootstrap 95% confidence intervals for the key RMSE differences across all 3,520 test days. CINN vs. MLP: +0.8 bps [0.6, 1.0]; CINN vs. SSVI: −12.3 bps [−13.1, −11.5]; CINN vs. SVI: −22.9 bps [−24.2,

−21.6]. All differences are statistically significant at $p < 0.001$ (paired bootstrap, 10,000 resamples). The CINN-MLP accuracy gap of 0.8 bps (4.3%) is small but genuine—it is the cost of arbitrage-freeness, not noise.

10 Conclusion

We have shown that the tradeoff between flexible interpolation and empirical arbitrage-freeness can be made favorable—4.3% accuracy loss for near-zero exploitable arbitrage at evaluation grid points—though it cannot be eliminated entirely without hard-constraint methods like Andreasen-Huge.

The core contribution is engineering, not theory: we show that penalty-based constrained optimization—either fixed penalty with warmup scheduling or augmented Lagrangian with adaptive dual variables—is sufficient to enforce known no-arbitrage conditions on neural volatility surfaces, even under realistic structured sparsity where both parametric and unconstrained neural methods fail. We formalize the constrained optimization problem (Eq. 3), implement an ALM solution (Eq. 9), and state convergence as an empirical conjecture (Conjecture 4.6.3) that we document but do not prove.

Main empirical findings:

- Under 60% uniform strike dropout, CINN achieves 0.04 bps mean EPP (0.12 bps worst case) at evaluation grid points vs 3.21 bps for per-slice SVI, 2.14 bps for jointly-calibrated SSVI, and 0.00 bps for Andreasen-Huge (which pays a 55% RMSE penalty for its hard guarantee)
- Soft penalties ($\lambda_{\text{but}} \leq 0.1$) are insufficient; they reduce but do not eliminate exploitable arbitrage. The Pareto frontier across 9 penalty configurations (Table 2) is smooth with no cliff-edge behavior
- Constrained training *reduces* initialization variance (RMSE std 0.7 vs 2.1 bps), through capacity restriction rather than generalization-improving regularization
- The penalty mechanism, not the specific architecture, is the primary driver of arbitrage-freeness—all tested architectures achieve near-zero EPP under the same penalty regime (Table 1)
- The warmup-then-ramp schedule is essential: skipping warmup causes 40% of seeds to converge to trivially flat solutions (Table 3)

Honest limitations:

- No theoretical guarantee of constraint satisfaction; it works empirically at discrete evaluation points
- 1.7× slower than unconstrained methods (412ms vs 245ms end-to-end)
- Requires warmup scheduling to avoid gradient domination (Table 3)
- Sparsity model uses uniform random dropout; real-world sparsity is structured (liquidity-dependent)
- Wing behavior beyond training moneyness range ($|k| > 0.25$) is not guaranteed to satisfy Roger-Lee bounds

- Beyond SPX and three mega-cap single-name equities, generalization to FX surfaces (different smile dynamics), small-cap equities (<5 strikes), or commodity options (seasonal term structure) is untested

When to use this: End-of-day risk reporting under sparse grids, illiquid markets, regulatory requirements for arbitrage-freeness. **When not to:** Latency-critical applications (<100ms budget), clean high-frequency data where SSVI suffices, wing extrapolation without blending.

Practical impact: For SPX options under uniform sparsity conditions (40–60% missing strikes), CINN reduces arbitrage to <0.05 bps at grid points while maintaining RMSE within 5% of unconstrained MLPs and 35% lower RMSE than Andreasen-Huge. On genuinely sparse single-name equities (AAPL, TSLA, JPM with 8–12 strikes/expiry), the same penalty configuration achieves near-zero EPP without retuning (Table 8).

Future directions:

- **ALM convergence theory:** Prove Conjecture 4.6.3—that ALM converges to a KKT point for the non-convex volatility surface problem under appropriate constraint qualification (LICQ or MFCQ). This requires establishing regularity of the Breeden-Litzenberger density coupling.
- **Enhanced parametric baselines:** Quasi-explicit SVI [De Marco and Martini, 2009] with no-arbitrage parameter constraints [Martini and Mingone, 2022]; piecewise-linear Andreasen-Huge; eSSVI [Hendriks and Martini, 2019]
- **Constraint-by-construction comparison:** Quantitative benchmark against input convex networks [Amos et al., 2017] for hard butterfly guarantees
- **Genuinely thin markets:** Small-cap equities (3–5 strikes, wide spreads), commodity options (seasonal gaps), EM FX
- **Extension to stochastic volatility:** Heston, rough Bergomi surfaces; online learning for real-time updates

Code and Data Availability

The CINN implementation (PyTorch) including both fixed-penalty and ALM training, the structured sparsity dropout model, baseline calibration code (SVI, SSVI, SABR, Andreasen-Huge), EPP computation, and scripts to reproduce all tables are available at <https://github.com/diegourdaneta/OptionsLab>. The figure generation scripts are included; figures in this paper were produced using `docs/research/generate_paper_figures.py`. Raw option data is from OptionMetrics IvyDB and cannot be redistributed due to licensing; researchers with OptionMetrics access can reproduce all results using the provided data loading pipeline. Pre-trained model weights are not included (models are calibrated per-surface, so there are no persistent weights to share).

A Counterexample: Non-Convexity

Example A.1. Consider SVI smiles:

$$w_1(k) = 0.04 + 0.1 \left(-0.5k + \sqrt{k^2 + 0.01} \right) \quad (14)$$

$$w_2(k) = 0.04 + 0.3 \left(0.5k + \sqrt{k^2 + 0.04} \right) \quad (15)$$

Both are admissible. Their average $w_{0.5} = 0.5w_1 + 0.5w_2$ has $g(k = 0.15) \approx -0.003 < 0$. Hence $\mathcal{A}$ is not convex.

B Why Not Projection?

A natural question: why not project onto the admissible set after training?

Answer: Because $\mathcal{A}$ is non-convex, no unique projection exists. Even if we could define one, computing $\Pi_{\mathcal{A}}(w)$ requires solving a constrained optimization problem *per surface*—precisely what we’re trying to avoid.

Some alternatives we considered:

- **Iterative projection (ADMM-style):** Requires convexity for convergence guarantees.
- **Barrier methods:** Numerically unstable near constraint boundaries.
- **Architectures with built-in constraints:** E.g., integrating a positive function to guarantee monotonicity. These underfit in practice because they over-regularize.

Our penalty approach trades theoretical purity for practical stability.

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